

PEI JIA DENG

Seattle, WA | pdeng2@uw.edu | github.com/DDPPJJ876

Education

University of Washington, Seattle

Expected 2029

B.S. in Computer Science, Paul G. Allen School of Computer Science & Engineering

Research Interests

Computer vision; multimodal learning; explainable and human-centered AI; interactive systems.

Research Experience & Projects

Facial Expression Recognition & Explainable AI

2025–Present

Independent research: paper reproduction and extensions

- Reproduced the authors' FER + XAI pipeline to study alignment between model explanations and facial regions defined by the Facial Action Coding System (FACS).
- Trained and evaluated **12 CNNs** with **5-fold cross-validation**; tracked checkpoints and evaluation results.
- Reproduced **SLIC** + **LIME** attribution maps to inspect regions driving facial-expression predictions.
- Found limited FACS alignment; designed **IoU evaluation** and attention-alignment training objectives.

Reality-Shake: Multimodal Brain–Computer Interaction

2025

Interactive system prototype

- Built an Arduino–Python–Unity pipeline to process BCI attention data and trigger real-time game events.
- Created a role-playing prototype in which users' cognitive state influences actions and task progression.

Multimodal Affective Interaction in Games

2025–Present

Independent exploratory research

- Exploring facial, behavioral, and sensor signals to adapt gameplay and non-player character (NPC) behavior; investigating robustness to noise, missing modalities, and real-time latency.

Biocompatible Brain–Computer Interfaces

2024

Literature review & conceptual research proposal; Advisor: Guanxiang Heng

- Co-authored a literature review on implanted BCIs, examining glial responses, electrode–tissue interactions, and long-term neural signal stability.
- Synthesized research on iPSC-derived neurons, gene editing, and cell adhesion to propose a neuron-coated electrode interface and outline evaluation of cell adhesion and neural signal quality.

Selected Competition

Kaggle Bronze Medal — CMI: Detect Behavior with Sensor Data

Sep. 2025

- Ranked **179th of 2,667 teams (top 6.7%)**; developed behavior-recognition models for noisy, multimodal wearable-sensor time-series data.

Additional Engineering Experience

DIGO

2026–Present

Founder & Developer

- Developing a demand-driven apparel platform with **Next.js**, **TypeScript**, **Supabase**, and **Stripe**, covering product presentation, reservations, order flows, and payment integration.
- Exploring digital apparel representations linking garment and material attributes to demand and production.

FIRST Robotics Competition (FRC), Team 8214, China

2021–2025

- Four years on the team; **mentor in Grade 12**; contributed to **scouting software development**.

Technical Skills

Languages: Python, Java, R, JavaScript/TypeScript

ML & Analysis: TensorFlow/Keras, CNNs, LIME, SLIC superpixels, cross-validation, Jupyter

Systems & Tools: Unity, Arduino, Git/GitHub, Next.js, Supabase, Stripe, Vercel